

Lecture 14: Extensions to Loglinear Models

MATH3823 Generalised Linear Models

Richard P Mann

MATH3823 Generalised Linear Models

Course notes: Chapter 7

www.richardpmann.com/MATH3823

The Problem: Fixed Marginals

In many studies, some totals are fixed by design:

- Clinical trial: Fixed number per treatment group
- Case-control study: Fixed cases and controls
- Survey: Fixed sample size

When totals are fixed:

- Cell counts are *not* independent Poisson
- The standard Poisson model is theoretically incorrect
- But we can still use Poisson GLM.

Example: Fixed Total

Melanoma study: Total of 400 patients (fixed).

Questions:

- Can we still use Poisson loglinear models?
- What distribution do the counts actually follow?
- How do we interpret the results?

Answer: The counts follow a *multinomial* distribution, but Poisson gives the same parameter estimates.

Proposition

If $Y_1, \dots, Y_k$ are independent $\text{Poisson}(\lambda_i)$, then conditional on $Y_+ = n$:

$$(Y_1, \dots, Y_k) \mid Y_+ = n \sim \text{Multinomial} \left(n; \frac{\lambda_1}{\lambda_+}, \dots, \frac{\lambda_k}{\lambda_+} \right)$$

Key insight: Conditioning independent Poissons on their sum gives a multinomial.

Proposition

If $Y_1 \sim \text{Poisson}(\lambda_1)$ and $Y_2 \sim \text{Poisson}(\lambda_2)$ independently, then:

$$Y_1 \mid (Y_1 + Y_2 = n) \sim \text{Binomial} \left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2} \right)$$

Application: In a 2×2 table with fixed row totals, the row distributions are binomial.

Product-Multinomial Distribution

When row totals y_{i+} are fixed:

Each row follows an independent multinomial:

$$(Y_{i1}, \dots, Y_{ic}) \sim \text{Multinomial}(y_{i+}; \pi_{i1}, \dots, \pi_{ic})$$

This is called a product-multinomial distribution.

The joint distribution is the product of row-wise multinomials.

The Key Theorem

Theorem

Tables with fixed margin sums can be analyzed using a multinomial or product-multinomial model as though they were independent Poisson models, provided terms corresponding to the fixed margins are included in the model.

What this means:

- Fixed total $y_{++} \Rightarrow$ include μ (intercept)
- Fixed row totals $y_{i+} \Rightarrow$ include row main effects α_i
- Fixed column totals $y_{+j} \Rightarrow$ include column main effects β_j

Why Does This Work?

Mathematical reason:

When we condition on fixed margins, the parameters for those margins “absorb” the constraint. The remaining parameters (e.g., interactions) have the same MLEs whether we use:

- Poisson likelihood
- Multinomial/product-multinomial likelihood

Practical implication:

We can use `glm(..., family = poisson)` even when the Poisson assumption is technically violated.

Application: Flu Vaccine Trial

Data: 73 participants, fixed group sizes

Group	Low	Moderate	High	Total
Placebo	25	8	5	38
Vaccine	6	18	11	35
Total	31	26	16	73

Row totals fixed: 38 placebo, 35 vaccine (by design).

Flu Data: R Analysis

```
# Create data
flu <- data.frame(
  Group = factor(rep(c("Placebo", "Vaccine"),
                    each = 3)),
  Response = factor(rep(c("Low", "Mod", "High"), 2),
                    levels = c("Low", "Mod", "High")),
  Count = c(25, 8, 5, 6, 18, 11)
)

# Independence model (must include Group for fixed rows)
model <- glm(Count ~ Group + Response,
             family = poisson, data = flu)
summary(model)
```

Flu Data: Results

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.6461	0.1796	14.729	< 2e-16 ***
GroupVaccine	-0.0816	0.2399	-0.340	0.734
ResponseMod	-0.1757	0.2721	-0.646	0.518
ResponseHigh	-0.6614	0.3164	-2.091	0.037 *

Residual deviance: 18.643 on 2 degrees of freedom

Test of independence:

- Deviance: 18.64 on 2 df
- p -value < 0.001
- Strong evidence of association between group and response

Guidelines for Fixed Margins

Two-way table:

Fixed	Include in model
Total only (y_{++})	Intercept μ
Row totals (y_{i+})	Row effects α_i
Column totals (y_{+j})	Column effects β_j
Both margins	Both α_i and β_j

Three-way table: Similar principle — include main effects and interactions corresponding to fixed margins.

1. Small expected counts:

- χ^2 approximation may be poor
- Rule of thumb: Expected counts ≥ 5
- Consider exact tests or combining categories

2. Overdispersion:

- If deviance $\gg$ df, consider `quasipoisson`
- Or investigate causes (missing covariates, clustering)

3. Sparse tables:

- Many zeros cause problems
- May need specialized methods

Key points:

- Fixed marginals $\Rightarrow$ counts are not independent Poisson
- Conditioning Poisson on sum gives multinomial
- Product-multinomial when row totals are fixed
- **Key theorem:** Can use Poisson GLM if fixed-margin terms included
- Include row effects if row totals are fixed
- Include column effects if column totals are fixed
- Parameter estimates for interactions are the same.

Next lecture: Course revision and summary.